

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intellegence

Practical Task– IV

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Class:TYIT	Batch:2
Date of Assignment: 25/07/26	Date/Time of Submission:25/07/26

PRACTICAL TASK

Q.1 In a college:

- 20% of students know Python.
- 80% of students know Java.
- 90% of Python students get placed.
- 40% of Java students get placed.

A randomly selected student is placed.

Find:

- Probability that the student knows Python.
- Probability that the student knows Java.
- Probability that a placed student knows Python.

Code:-

Given probabilities

P_Python = 0.20

P_Java = 0.80

P_Placed_Python = 0.90

P_Placed_Java = 0.40

Total probability of placement

P_Placed = (P_Placed_Python * P_Python) + (P_Placed_Java * P_Java)

Bayes theorem

P_Python_Placed = (P_Placed_Python * P_Python) / P_Placed

print("Ishita Patel")

print("Probability student knows Python:", P_Python)

print("Probability student knows Java:", P_Java)

print("Probability that a placed student knows Python:", round(P_Python_Placed, 2))

Output:-

```
Ishita Patel
Probability student knows Python: 0.2
Probability student knows Java: 0.8
Probability that a placed student knows Python: 0.36
```

Q.2 In a city:

- 85% of taxis are Green.
- 15% are Blue.
- A witness correctly identifies the taxi color 80% of the time.

A witness claims the taxi involved in an accident was Blue. What is the probability that the taxi was actually Blue?

Code:-

```
# Given probabilities
```

```
P_Blue = 0.15
```

```
P_Green = 0.85
```

```
P_SaysBlue_given_Blue = 0.80
```

```
P_SaysBlue_given_Green = 0.20
```

```
# Total probability
```

```
P_SaysBlue = (P_SaysBlue_given_Blue * P_Blue) + (P_SaysBlue_given_Green * P_Green)
```

```
# Bayes theorem
```

```
P_Blue_given_SaysBlue = (P_SaysBlue_given_Blue * P_Blue) / P_SaysBlue
```

```
print("Ishita Patel")
```

```
print("Probability taxi was actually Blue:", round(P_Blue_given_SaysBlue, 3))
```

Output:-

```
Ishita Patel
Probability taxi was actually Blue: 0.414
```